Minimization of DFA:

1.

Step I. All states are reachable.

State Transition Table (DFA)

	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_1	q_3
q_2	q_1	q_2
q_3	q_1	q_4
$* q_4$	q_1	q_2

Step II.

$\pi_0 = \{ \{q_0, q_1, q_2, q_3\}, \{q_4\} \}$

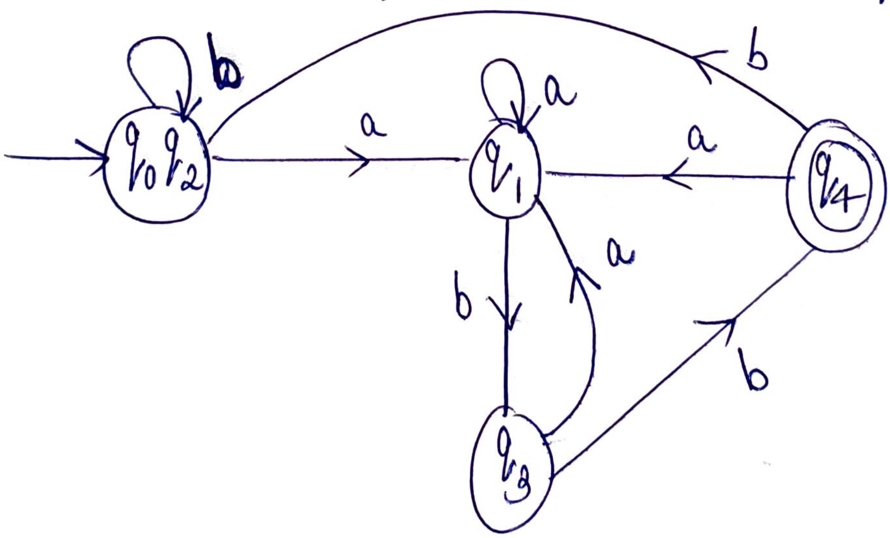
$\pi_1 = \{ \{q_0, q_1, q_2\}, \{q_3\}, \{q_4\} \}$

$\pi_2 = \{ \{q_0, q_2\}, \{q_1\}, \{q_3\}, \{q_4\} \}$

$\pi_3 = \{ \{q_0, q_2\}, \{q_1\}, \{q_3\}, \{q_4\} \}$

$\therefore \quad \pi_3 = \pi_2$

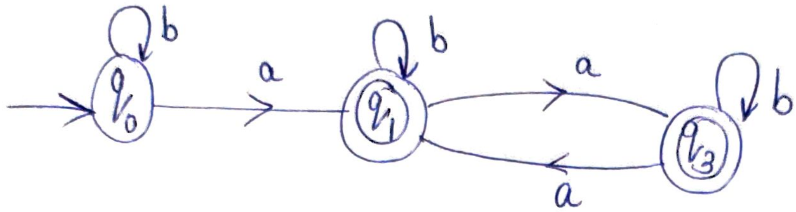
Initial state : $\{q_0 q_2\}$, Final state : $\{q_4\}$



Minimized DFA

(2)

Step (I) Since state q_2 is unreachable from initial state. Hence it can be removed.



Step (II)

	a	b
$\rightarrow q_0$	q_1	q_0
$*q_1$	q_3	q_1
$*q_3$	q_1	q_3

$\pi_0 = \{ \{q_0\}, \{q_1, q_3\} \}$

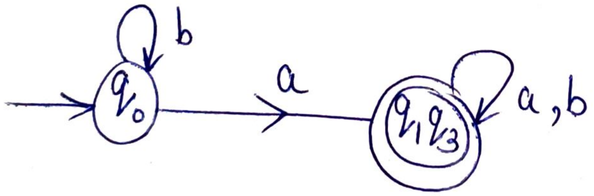
$\pi_1 = \{ \{q_0\}, \{q_1, q_3\} \}$

$\therefore \quad \pi_1 = \pi_0$

Initial state : $\{q_0\}$

Final state : $\{q_1 q_3\}$

Step (III) DFA :



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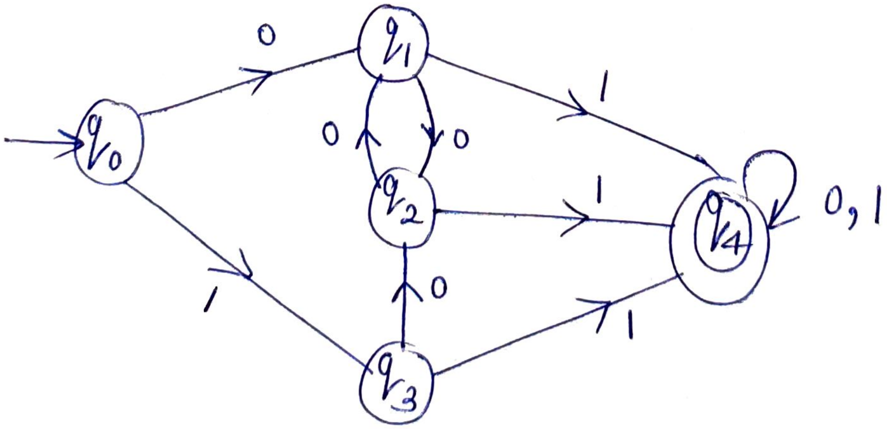
(3)



Step (I) All states are reachable.

Step (II)

	0	1
$\rightarrow q_0$	q_1	q_3
q_1	q_2	q_4
q_2	q_1	q_4
q_3	q_2	q_4
$* q_4$	q_4	q_4

$\pi_0 = \{ \{q_0, q_1, q_2, q_3\}, \{q_4\} \}$

$\pi_1 = \{ \{q_0\}, \{q_1, q_2, q_3\}, \{q_4\} \}$

$\pi_2 = \{ \{q_0\}, \{q_1, q_2, q_3\}, \{q_4\} \}$

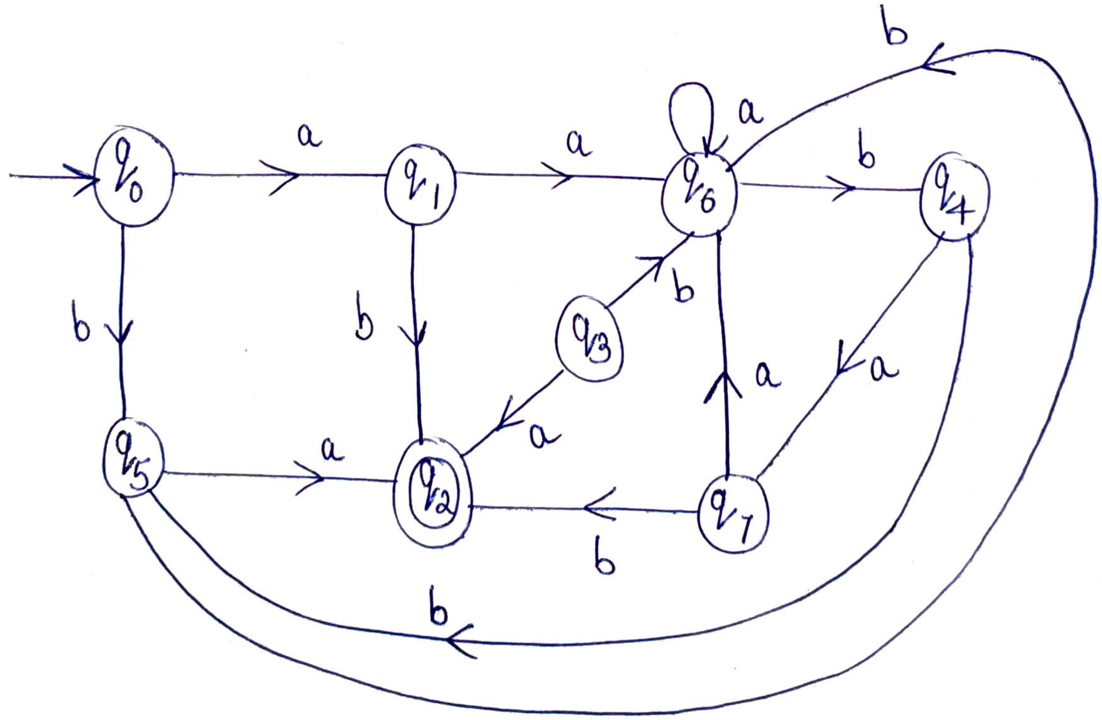
$\therefore \quad \pi_2 = \pi_1$

Initial state : $\{q_0\}$

Final state : $\{q_4\}$

Step (III): Minimized DFA :

(4.)



Step (I) Since state q_3 is unreachable from initial state. Hence we can remove it.

Step (II):

	a	b
$\rightarrow q_0$	q_1	q_5
q_1	q_6	q_2
$*q_2$	q_0	q_2
q_4	q_7	q_5
q_5	q_2	q_6
q_6	q_6	q_4
q_7	q_6	q_2

$\pi_0 = \{ \{q_0, q_1, q_4, q_5, q_6, q_7\}, \{q_2\} \}$

$\pi_1 = \{ \{q_0, q_4, q_6\}, \{q_1, q_7\}, \{q_5\}, \{q_2\} \}$

$\pi_2 = \{ \{q_0, q_4\}, \{q_1, q_7\}, \{q_5\}, \{q_6\}, \{q_2\} \}$

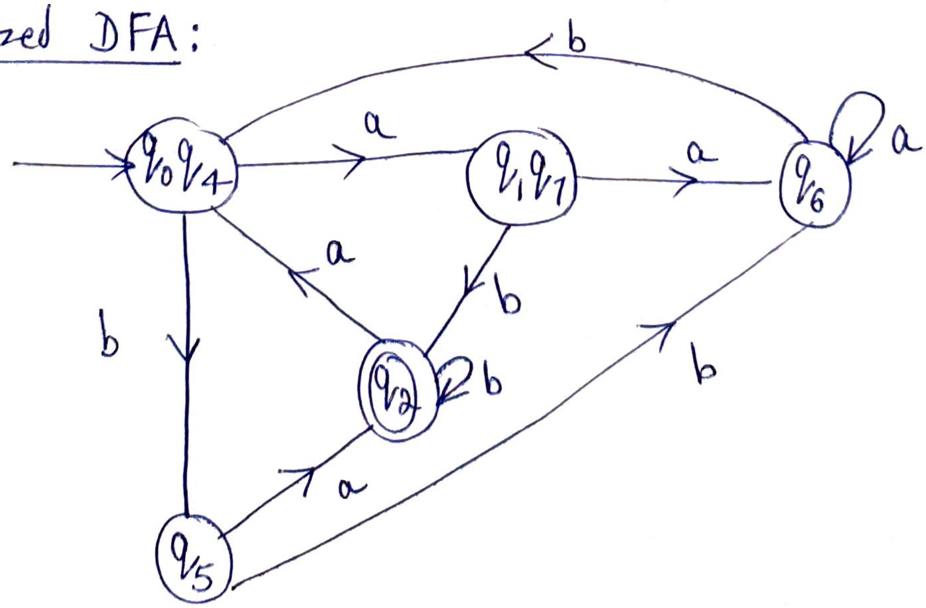
$\pi_3 = \{ \{q_0, q_4\}, \{q_1, q_7\}, \{q_5\}, \{q_6\}, \{q_2\} \}$

$\therefore \quad \pi_3 = \pi_2$

Initial state : $\{q_0 q_4\}$

Final state : $\{q_2\}$

Step (III): Minimized DFA:



(5.)

Step (I.) All states are reachable.

State Transition Table (DFA)

	0	1
$\rightarrow q_0^+$	q_1	q_3
q_1	q_4	q_2
q_2^+	q_1	q_5
q_3	q_0	q_4
q_4	q_4	q_4
q_5	q_2	q_4

Step (II.)

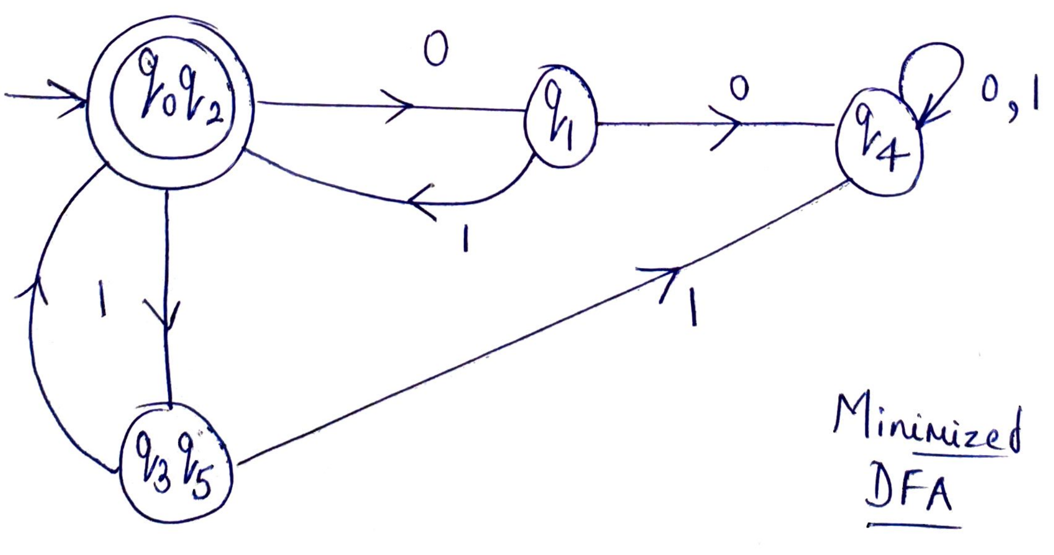
$\pi_0 = \{ \{q_1, q_3, q_4, q_5\}, \{q_0, q_2\} \}$

$\pi_1 = \{ \{q_1\}, \{q_3, q_5\}, \{q_4\}, \{q_0, q_2\} \}$

$\pi_2 = \{ \{q_1\}, \{q_3, q_5\}, \{q_4\}, \{q_0, q_2\} \}$

$\therefore \pi_2 = \pi_1$

Initial state: $\{q_0 q_2\}$, Final state: $\{q_0 q_2\}$



Minimized DFA

6.

Step I. All states are reachable.

Step II.

	0	1
$\rightarrow q_0$	q_3	q_1
$*q_1$	q_2	q_5
$*q_2$	q_2	q_5
q_3	q_0	q_4
$*q_4$	q_2	q_5
q_5	q_5	q_5

$\pi_0 = \{ \{q_0, q_3, q_5\}, \{q_1, q_2, q_4\} \}$

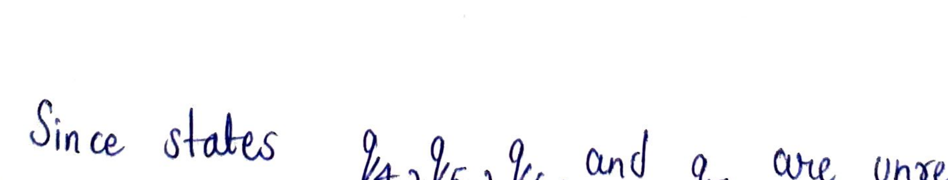
$\pi_1 = \{ \{q_0, q_3\}, \{q_5\}, \{q_1, q_2, q_4\} \}$

$\pi_2 = \{ \{q_0, q_3\}, \{q_5\}, \{q_1, q_2, q_4\} \}$

$\therefore \pi_2 = \pi_1$

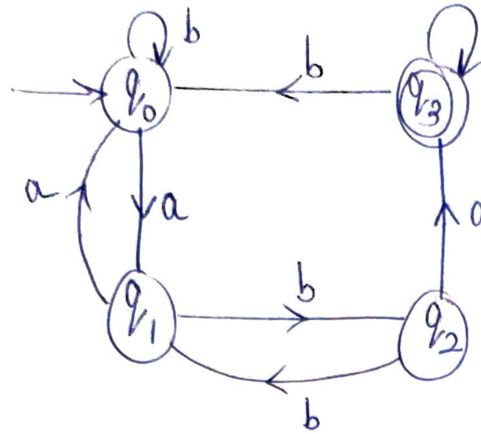
[Initial state: $\{q_0 q_3\}$, Final state: $\{q_1 q_2 q_4\}$]

Step III. Minimized DFA:



7.

Step I. Since states q_4, q_5, q_6 and q_7 are unreachable from initial state of given DFA. Hence all q_4, q_5, q_6 & q_7 can be removed.



Step (II)

	a	b
$\to q_0$	q_1	q_0
q_1	q_0	q_2
q_2	q_3	q_1
$*q_3$	q_3	q_0

$\pi_0 = \{ \{q_0, q_1, q_2\}, \{q_3\} \}$

$\pi_1 = \{ \{q_0, q_1\}, \{q_2\}, \{q_3\} \}$

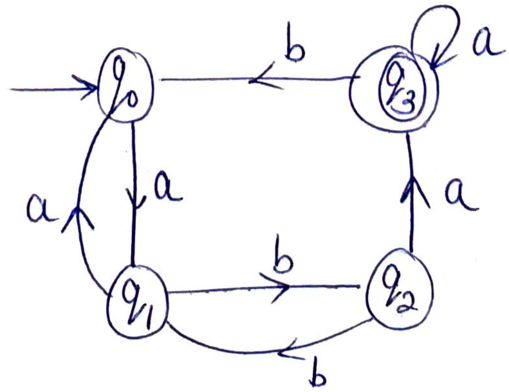
$\pi_2 = \{ \{q_0\}, \{q_1\}, \{q_2\}, \{q_3\} \}$

$\pi_3 = \{ \{q_0\}, \{q_1\}, \{q_2\}, \{q_3\} \}$

$\therefore \pi_3 = \pi_2$

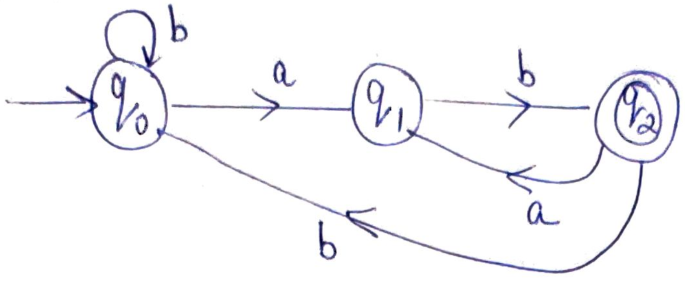
[Initial state : $\{q_0\}$
Final state : $\{q_3\}$]

Step (III) Minimized DFA :

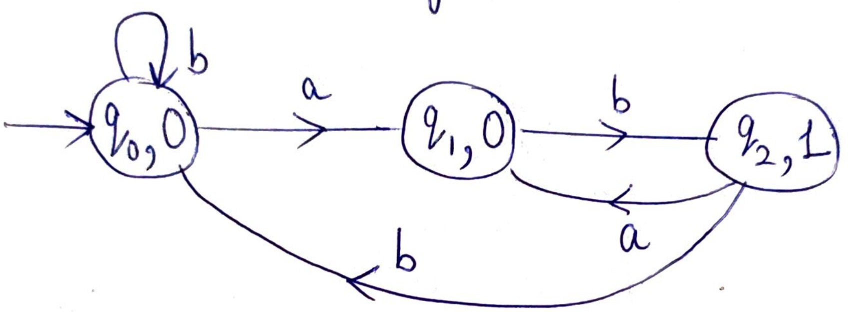


Moore & Mealy Machine:

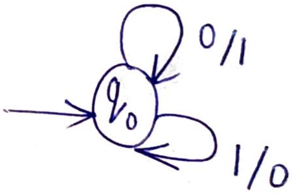
① DFA: all strings ending with "ab".



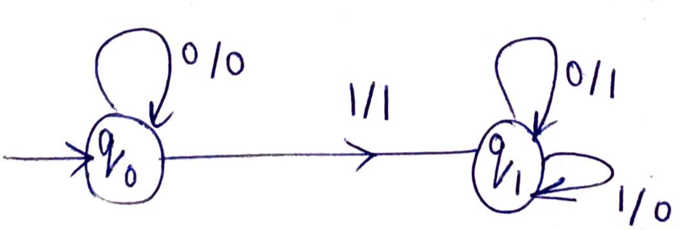
Moore Machine: which counts the occurrence of substring "ab".



② 1's Complement Mealy Machine:

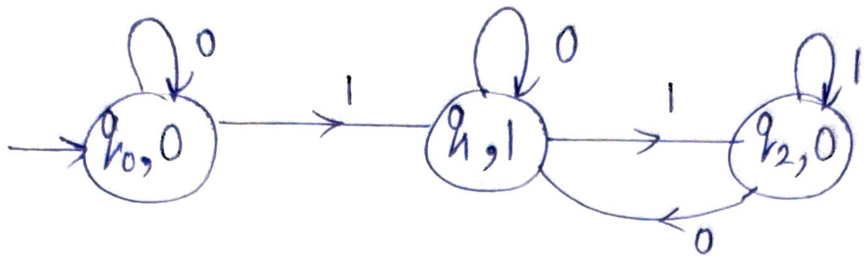


③ 2's Complement Mealy Machine:

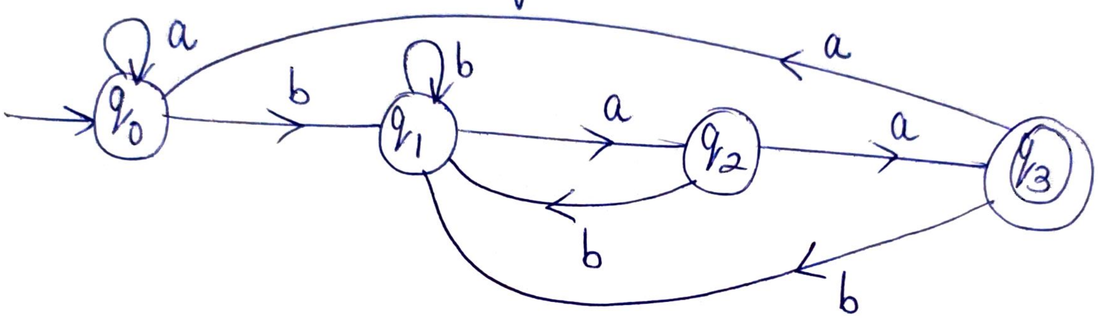


Note: The input string should be read from rightmost bit.

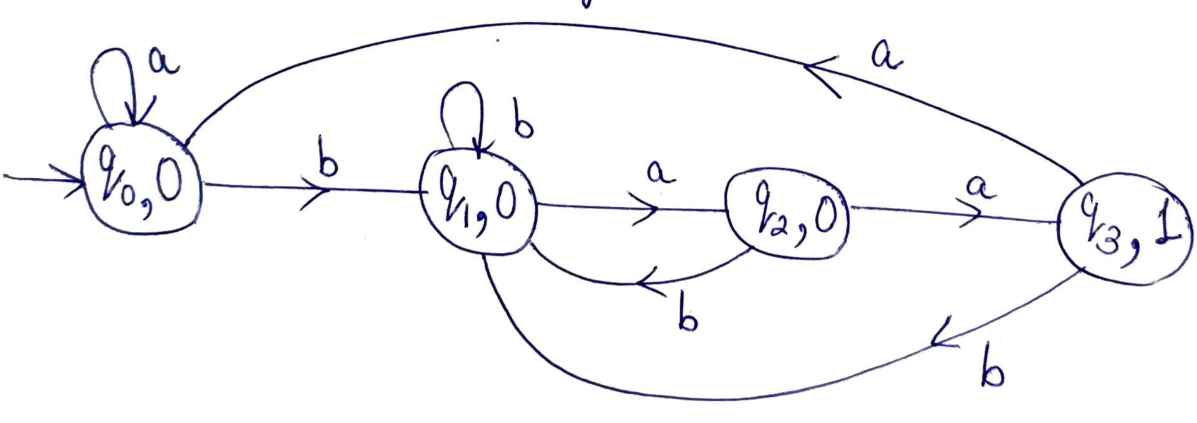
(4.) 2's Complement Moore Machine:



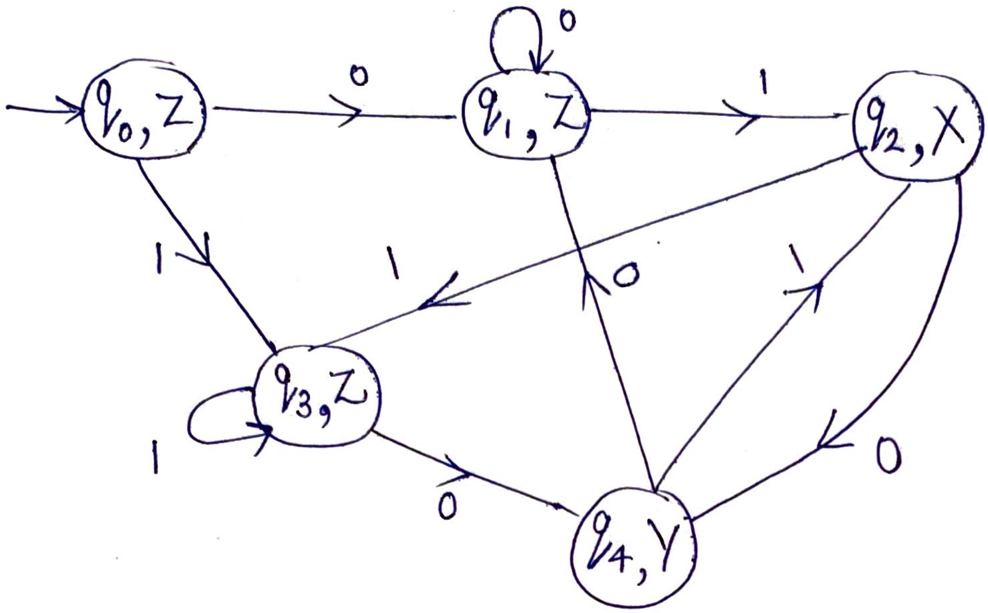
(5.) DFA: all strings ending with "baa".



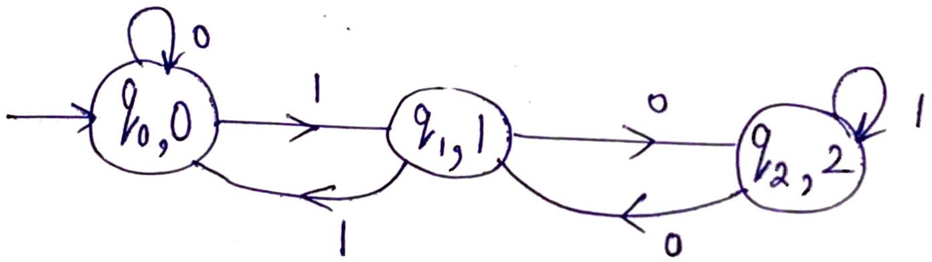
Moore Machine: which prints 1 for each occurrence of substring "baa".



6. Moore Machine:

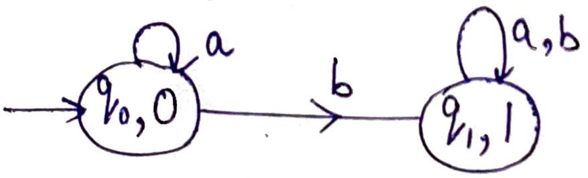


7.



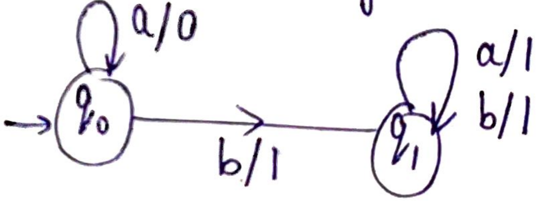
8.

Moore



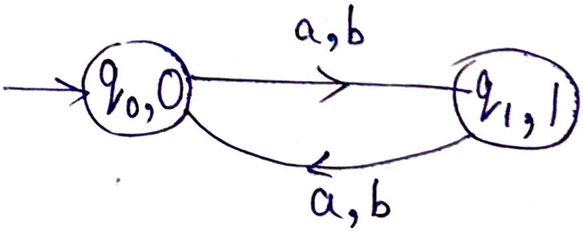
⇒

Mealy



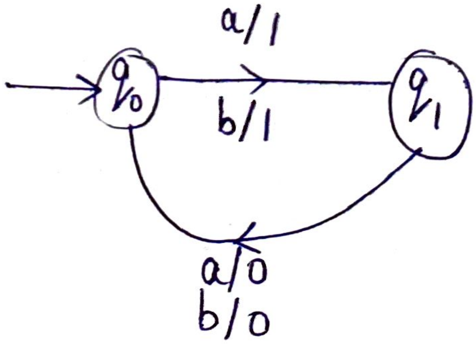
9.

Moore



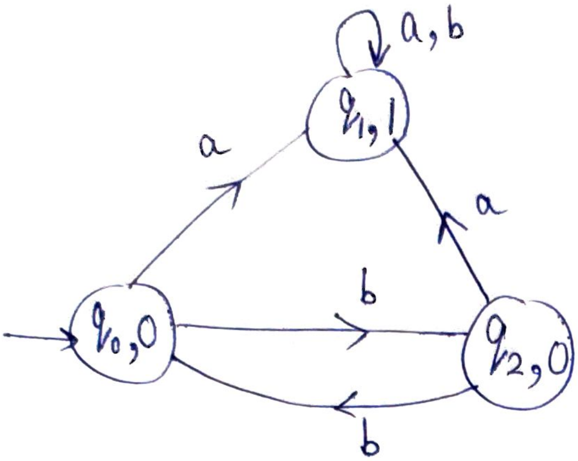
⇒

Mealy



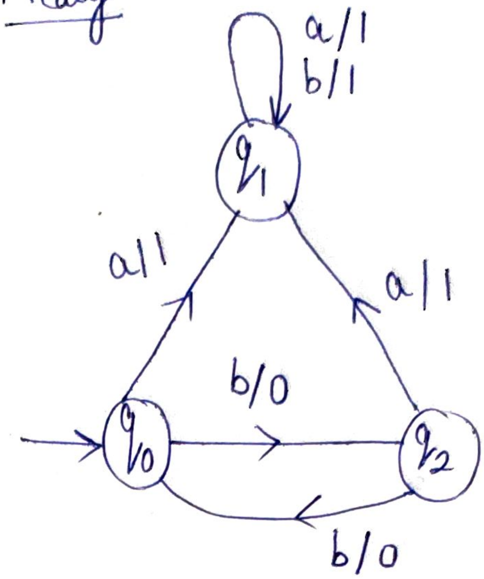
10.

Moore



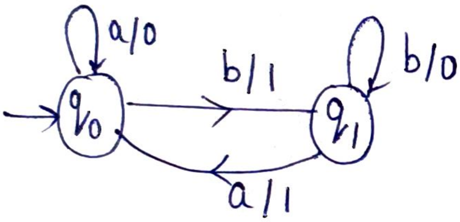
⇒

Mealy



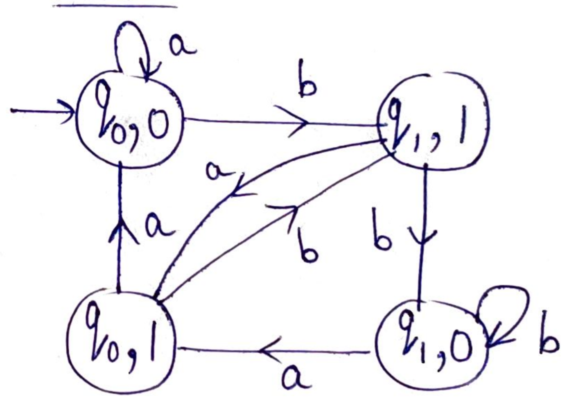
11.

Mealy

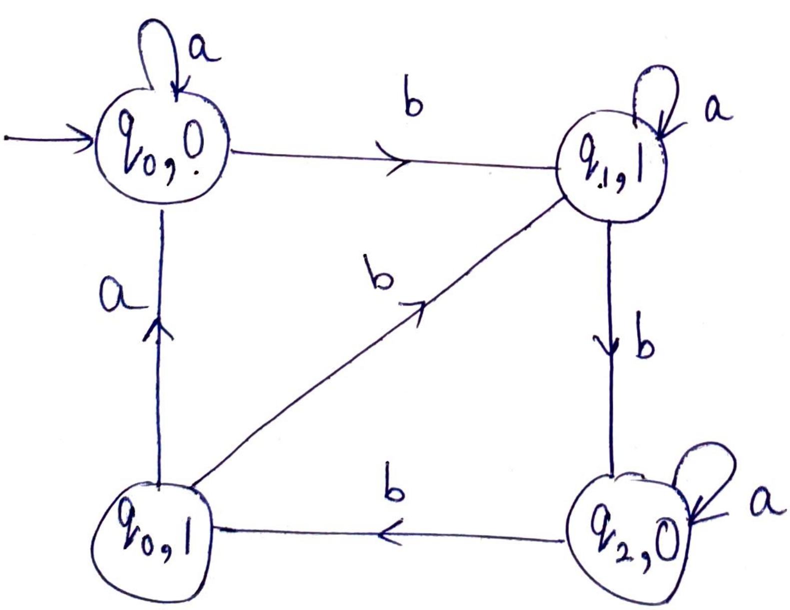


⇒

Moore

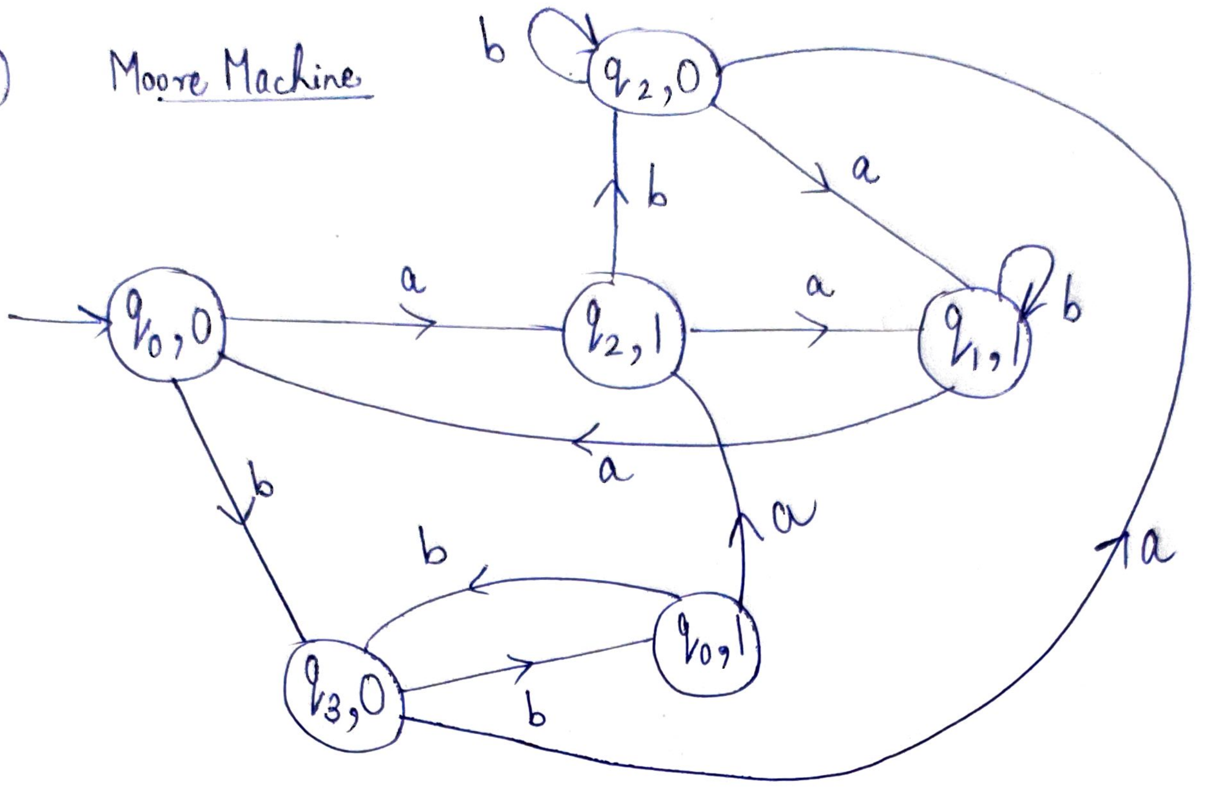


12.



Moore

(13.) Moore Machine



(14.)

Moore Machine